

Umer Irfan

Software Engineering · Voice Agents · Applied Machine Learning
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PROFESSIONAL EXPERIENCE

- AI/ML Engineer | [Streaming Voice Agent](#) Aug 2024 to Mar 2025
Sales demo prototype under [Asif Mufti](#). [Carl Zeiss](#) was a prospective customer.
 - Built a **real-time sales voice agent** using Node.js, WebSockets and a Python backend on DigitalOcean. Combined WebRTC VAD, Whisper transcription and GPT-4 responses with configurable sales personas.
 - Added PyAudio streaming, session handling and MySQL call records. Tuned buffering for spoken responses in **under four seconds**.
- React Native Engineer | [Project X](#) Nov 2024 to Jun 2025
 - Built a smart-camera Android MVP with **device onboarding**, Firebase authentication, Google sign-in and notifications so users could pair cameras and receive alerts in one app.
 - Worked on ESP32 camera firmware for OTA update fetching and MQTT telemetry. [Firmware case study](#).
- DevOps Engineer | [U.n.I Social App](#) May to Jul 2024
 - Built **GitHub Actions deployment workflows** for container delivery to EC2. Configured NGINX routing to a higher-capacity instance, automating releases and handling load.

APPLIED AI AND RESEARCH

- [Kissan-Dost](#) | Urdu Conversational Agriculture IEEE DCOSS-IoT 2026
 - Co-developed an **Urdu conversational AI interface** that gave farmers field-grounded advice over WhatsApp voice and text. Built a multi-hop ESP-NOW sensing mesh and ESP32 gateway.
 - Two 45-day deployments supported a five-person pilot with near-daily chatbot use and irrigation decisions informed by its advice. Co-author. [Paper](#).
- [PARDA](#) | Audio Privacy for Smart Glasses PerCom 2027 submission
 - Applied **SEAL-style redaction** (adversarial sensitive-content anonymisation) to conversation context. Distilled a retrieval-aware adversary and anonymizer into a **2B model** using 717 CANDOR conversations.
 - On 143 held-out conversations, model-judged privacy leakage fell from 0.502 to 0.378 while utility rose from 0.875 to 0.900. First author. [Project](#).
- [LLM-Enhanced Wearables](#) | Health Guidance arXiv 2026
 - Developed a **\$12.72 screenless wearable** and plain-language WhatsApp feedback. In a 20-person, 96-hour study, mean self-rated comprehension rose from 3.05 to 3.60 out of 5. Co-author. [Paper](#).
- [WatchTower](#) | Network Anomaly Detection 2025
 - Built a transformer detector with 28 packet features and PROSER open-set recognition. Optimized INT8 inference with TVM on Raspberry Pi 5. Reported **3.82 ms inference** and 93.66% accuracy. [Project](#).

SELECTED PRODUCT AND SOFTWARE PROJECTS

- [Wellness Lamp](#) | React Native and ESP32 2026
 - Built a **React Native control app** for lighting themes, sound, BLE control and mood-based palettes. Synchronized ESP32 nodes so one app controlled multiple lamps. [Source](#).
- [Fintelligent](#) | Client Website 2026
 - Built a responsive [public site](#) in Next.js, React and TypeScript so prospective clients could find financing services and contact the firm.
- [Makers Lab Dashboard](#) | Full-Stack Tool 2024
 - Built a **React, Express and PostgreSQL** dashboard that brought equipment, staff, tasks and activity into one lab workspace. Added role-based sessions and Docker Compose packaging. [Source](#).
- [Violet Peer Tutoring](#) | Android and AI Prototype 2026
 - Built Android/Firebase scheduling screens and prototyped an **AI study assistant** and generated flashcards to support students between tutoring sessions.
- [PixelPacker](#) | Edge Neural Compression 2025 to 2026
 - Built a distilled INT8 CNN encoder with a **36-fold smaller memory footprint** and an Artix-7 acceleration path. Reported **over 10 times lower bitrate than JPEG** at comparable perceptual quality, enabling image transfer over constrained wildlife links.
- [Redis Cache](#) | TypeScript Service 2023
 - Added Redis caching to a PostgreSQL-backed TypeScript service. Measured **sub-10 ms cached reads** for faster repeated user queries in a local experiment. [Source](#).

TECHNICAL SKILLS

- Applications and services**
TypeScript · JavaScript · Python · React · React Native · Next.js · Node.js · Express · REST · WebSockets
- Data and deployment**
PostgreSQL · MySQL · Redis · Firebase · EC2 · Docker Compose · GitHub Actions · NGINX
- Applied ML and audio**
PyTorch · Whisper · WebRTC VAD · model distillation · retrieval · INT8 quantization · TVM
- Connected applications**
BLE · ESP32 · MQTT · OTA updates · Java · C · C++

EDUCATION

Lahore University of Management Sciences. BSc Computer Science. Minor in Computer Engineering. Sept 2023 to Jul 2027. CS GPA 3.707/4.00.

HONOURS AND AWARDS

- 1st Place.** [Pakistan Semiconductor Summit Hackathon](#) 2026. RISC-V RTL verification.
 - Dean's Honour List. LUMS. Three semesters.
- Best Project. Intel ISEF Pakistan 2019.
 - Best in Category. Teknofest 2020. Gaziantep, Turkey.
 - Team Pakistan. [Young Physicists Tournament](#), Beijing (2018).